FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2019

PAPER –2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

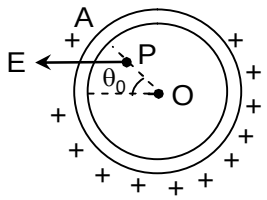
Sol. Using conservation of angular momentum about fixed charged.

$MV \sin 30R = Mv'.r \Rightarrow \frac{V}{2}R = \frac{V}{\sqrt{3}}r$

$r = \frac{\sqrt{3}}{2}R$

2. A

Sol. Charge on sphere will redistribute due to electric filed produced by thin plate. Electric field at any point (say P) due to charges on sphere is



$E = \frac{\phi}{2\varepsilon_{o0}}$ (so that field at P may become zero combinedly due to thin plate and sphere) Now due to sphere: $V_0 - V_A = (E\cos\theta_0)R$

$V_A = V_0 - E_r\cos\theta_0$

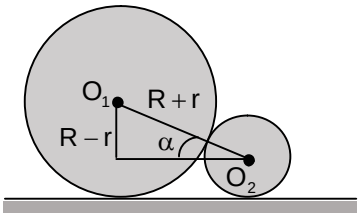
$= \frac{kQ}{R} - \frac{\sigma}{2\varepsilon_0}R\cos\left(\frac{\pi}{3}\right)$

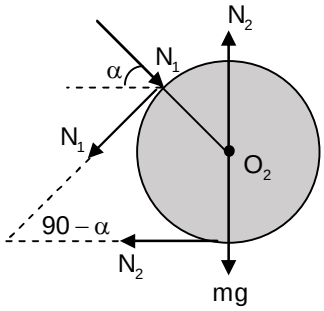
$= \frac{kQ}{R} - \frac{\sigma R}{4\varepsilon_0}$

[Here all the calculations are done only for the charges on spheres].

3. D

Sol. Figure shows the forces action on the smaller cylinder. N_1 is reaction between the two cylinders and N_2 the normal reaction between the smaller cylinder and the horizontal surface.





The conditions of equilibrium are

$$\sum F_x = N_1\cos\alpha - \mu N_2 - \mu N_1\cos(90-\alpha) = 0 \quad \text{....(i)}$$

Taking torque about O_2, centre of smaller sphere,

$$\sum \tau = \mu N_1 r - \mu N_2 r = 0$$

$$N_1 = N_2 \quad \text{....(ii)}$$

On substituting equations (ii) in (i), we obtain

$$\mu N_1 + \mu N_1 \sin\alpha = N_1\cos\alpha$$

$$\mu(1+\sin\alpha) = \cos\alpha \quad \text{....(iii)}$$

From figure (b), $\sin\alpha = \sqrt{\dfrac{R-r}{R+r}}, \cos\alpha = \dfrac{2\sqrt{Rr}}{R+r}$

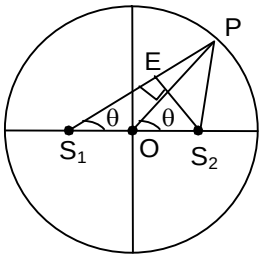
Now equations (iii) becomes; $\mu\left(1+\dfrac{R-r}{R+r}\right) = \dfrac{2\sqrt{Rr}}{R+r}\mu = \sqrt{\dfrac{r}{R}}$

Hence the required condition is $\mu \geq \sqrt{\dfrac{r}{R}}$.

4. A, B, C

Sol. The path different at point P is

$\Delta x = S_1E = 2D\cos\theta$



For maxima, $2D\cos\theta = mD$

$\therefore \cos\theta = \dfrac{m}{2}$

The possible value of m in first quadrant is 0, 1, 2.

Thus number of maxima in all four quadrants is eight.

For minima, $\Delta x = (2m-1)\dfrac{D}{2}$

$\cos\theta = \dfrac{(2m-1)}{4}$

Thus possible value of m in first quadrant is 1, 2
Thus, total number of minima in all four quadrant is eight.

5. A, D

Sol. $P_A = P + P_0 + \rho gh$ ....(i)

Apply Bernoulli's theorem at A and C and E

$$(P + P_0) + \rho gh = P_c + \rho\frac{V_c^2}{2}$$

$$= P_0 + \rho\frac{V_E^2}{2} \quad \text{....(ii)}$$

$$P_c = P_0 + \rho gh_1 \quad \text{....(iii)}$$

$$A_c V_c = A_E V_E$$

$$A_c V_c = \left(\frac{A_c}{2}\right) V_E \quad \text{....(iv)}$$

From eq. (i),

$$(P_0 + \rho gh_1) + \rho\frac{V_c^2}{2} = P_0 + \rho(2V_c)^2$$

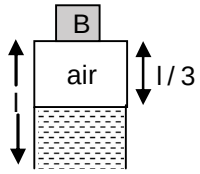
$$P + \rho gh = \rho g\left(\frac{7V_c^2}{2}\right)$$

$$v_c^2 = \frac{(P + \rho gh)}{4\rho}$$

Discharge rate $= A_c V_c = A\left(\frac{(P + \rho gh)}{4\rho}\right)^{1/2}$

6. A, C

Sol. The free body diagram of block + tube inclusive of water is as shown below



Since the block + tube system shown in figure is in equilibrium

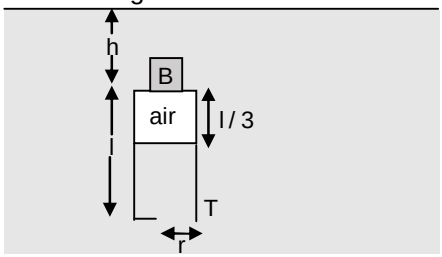
∴ Net weight of system = buoyant force

$$2d\,Vg + \pi r^2\frac{2l}{3}dg = (V + \pi r^2 l)dg$$

Where V is the volume of block B

$$(2d - d)V = \frac{\pi r^2 ld}{3} \text{ or } V = \frac{\pi r^2 l}{3}$$

The pressure of the air trapped inside the tube is same as pressure at point A in the water as shown in figure.



$$P = P_0 + \left(h + \frac{l}{3}\right)dg \quad P_a = P_0 + dg\left(h + \frac{l}{3}\right)$$

7. C, D

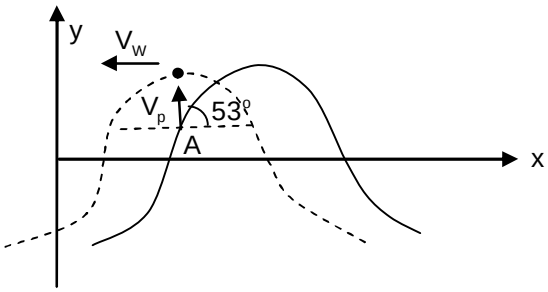
Sol. The temperature gradient (which causes the heat to flow) is only along the length. The temperature at any point of a given cross-section is the same and

so $Q_a = Q_b = 0$.

In the steady state, Q_1 is constant $\neq 0$

8. B, C

Sol. As shown in the curve, if wave is moving alogn $-x$ axis, V_P is positive.



$$\frac{V_p}{V_w} = -\tan\theta$$

$$\left|-\tan\theta\right| > 1 \qquad \therefore \quad V_P > V_W .$$

9. C

10. D

SECTION – C

11. 6

Sol. BE of X = 6A

Be of Y = 6A - 2 + 1 = 6A - 1

[Because absorption of energy decreases BE and release of energy increases BE] In Y nucleus there are A + 1 nucleus.

$$\therefore \frac{BE}{nuclon} = \frac{6A-1}{A+1}$$

12. 0

Sol. When switch S_2 is closed, due to symmetry no charge will flow through S_2.

13. 5

Sol. Let initial extension in spring is x_0 , then

$2kx_0\cos 45^o = mg\sin 45^o \Rightarrow 2kx_0 = mg$

Let after cutting right spring, acc. of ball is a_1 along the inclined plane and a_2 perpendicular to a_1

$$\Rightarrow \quad \frac{mg}{\sqrt{2}} - \frac{mg}{2}\frac{1}{\sqrt{2}} = ma_1 \Rightarrow a_1 = \frac{g}{2\sqrt{2}}$$

Then $mg\sin 45^o \Rightarrow kx_0\cos 45^o = ma_1$

$$kx_0\sin 45^o = ma_2 \Rightarrow a_2 = \frac{g}{2\sqrt{2}}$$

Net acceleration $a = \sqrt{a_1^2 + a_2^2} = \frac{g}{2} = 5\ m/s^2$

14. 6

Sol. Applying Bernoulli's theorem for above the plate (2) and below the plate (1), we have

$$p_1 = p_2 + \frac{1}{2}pv^2$$

$$\Rightarrow \quad p_1 - p_2 = \frac{1}{2}pv^2$$

Upward force is $(p_1 - p_2)A = \frac{1}{2}pv^2A$

$$\therefore \quad \frac{1}{2}pv^2A = mg$$

$$\Rightarrow \quad v = \sqrt{\frac{2mg}{pA}} = 6\sqrt{\frac{mg}{18pA}}$$

$$\Rightarrow \quad \gamma = 6$$

15. 4

Sol. Mass of air entering in 1 s is $m = \pi b^2 vp$

Change in momentum in 1 s $= mv - 0 = \pi b^2 pv^2$

Thus, $F = \pi b^2 pv^2$

Pressure due to this force is $\Delta p = \frac{F}{A} = \frac{\pi b^2 pv^2}{\pi b^2} = pv^2$

When this excess pressure equals to $\frac{4T}{R}$, the bubble gets separated.

$$\therefore \quad pv^2 = \frac{4T}{R} \Rightarrow R = \frac{4T}{pv^2}$$

So, $\beta = 4$

16. 5

Sol. The F.B.D. of wire PQ is the force due to surface tension $= F_{ST} = 2T \times 2AD\tan\theta$

$F_{ST} = (2T)\ 2AD\tan\theta$ $\qquad$ $F_{ST} = (2T)\ 2(AD + x)\tan\theta$

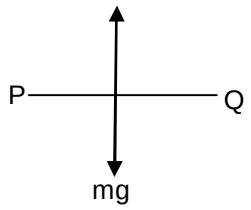


Figure (a)

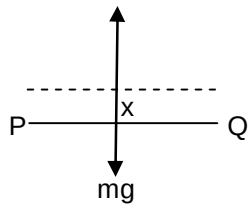


Figure (b)

For wire to be in equilibrium (Figure (a))

$4T\ AD\tan\theta = mg$

If the wire PQ is at a distance x below the mean position, the restoring force on the wire is (Figure (b))

$-ma = 4T\tan\theta(AD + x) - mg - 4T\tan\theta\ x$

Hence, the wire PQ executes SHM

$a = \frac{4T}{m}\tan\theta\ X$

Comparing with $a = -\omega^2 x$, we get

$\omega^2 = \frac{4T}{m}\tan\theta$

or $T = 2\pi\sqrt{\frac{m}{4T\tan\theta}} = 2\pi\sqrt{\frac{1\times10^{-3}}{4\times25\times10^{-3}}} = \frac{\pi}{5}s$

17. 6

Sol. In the figure, $\theta = 30^o$.

When displacement is each ball x, the extension in each spring is

$x' = x\cos\theta = x\frac{\sqrt{3}}{2}$

Each spring is connected to two balls.

So, net extension of each spring $= 2x' = \sqrt{3}x$

Force along the direction of displacement on each ball is

$F = k\sqrt{3}x\cos30^o + k\sqrt{3}x\cos30^o = 3kx$

$F = -3$ kx [negative because restoring in nature]

$\Rightarrow \quad a = \frac{F}{m} = -\frac{3k}{m}x \Rightarrow \omega = \sqrt{\frac{3k}{m}}$

$\Rightarrow \quad f = \frac{2}{2\pi}\sqrt{\frac{3k}{m}} = \frac{1}{2\pi}\sqrt{\frac{6k}{2m}} \Rightarrow \beta = 6$

18. 2

Sol. $a_{max} = \omega^2 A = g$

$\omega = \frac{2\pi v}{\lambda}, v = \sqrt{\frac{F}{\mu}}$

$\therefore \quad \omega^2 = \frac{4\pi^2 F}{\lambda^2}\times\frac{F}{\mu}$

$\therefore \quad a_{max} = \frac{4\pi^2 F}{\lambda^2\mu}.A_{min} = g$

$A_{min} = \frac{g\lambda^2\mu}{4\pi^2 F} = \frac{\lambda^2\mu}{4F} \qquad (\because \ \pi^2 = g)$

$A_{min} = \frac{(0.5)^2(0.1)}{4(3.125)}$

$= 2\times10^{-3}$ m $= 2$ mm

19. 4

Sol. Let the velocities of car 1 and car 2 be V_1 m / s and V_2 m / s

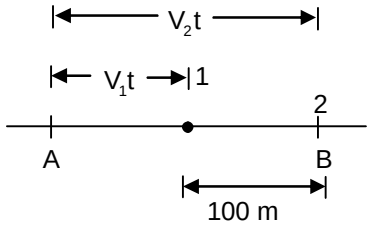
$\therefore$ Apparent frequencies of sound emitted by car 1 and car 2 as detected at end point are

$f_1 = f_0\frac{V}{V - V_1}, f_2 = f_0\frac{V}{V - V_2}$

$\Rightarrow \quad 330 = 300\frac{330}{330 - V_1}, 360 = 300\frac{330}{330 - V_2}$

$\Rightarrow \quad V_1 = 20$ m / s and $V_2 = 100$ m / s.

The distance between both the cars just when the 2nd car reach and point B (as shown in figure is)



$100\ m = V_2t - V_1t \Rightarrow t = 4\ sec.$

20. 1

Sol. $f = \dfrac{1}{2L}\sqrt{\dfrac{F}{\mu}}$

$f = c\sqrt{F}$

where constant $c = \dfrac{1}{2L\sqrt{\mu}}$

$$\frac{df}{dF} = c.\frac{1}{2\sqrt{F}}$$

or $$\frac{df}{dF} = \frac{1}{2L\sqrt{\mu}}.\frac{1}{2\sqrt{F}} = \frac{1}{2L\sqrt{\mu}}\frac{\sqrt{F}}{2F}$$

$$\frac{df}{dF} = \frac{1}{2L}\sqrt{\frac{F}{\mu}}.\frac{1}{2F} = \frac{f}{2F}$$

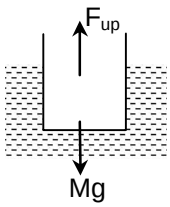
$$df = \frac{fdF}{2F} = \frac{(200)(1)}{2(100)} = 1\,s^{-1}$$

$\therefore$ Beat frequency = $1\,s^{-1}$

SECTION – D

21. 00002.67

Sol.



At equilibrium, $F_{up} = Mg$

22. 00015.70

Sol. $y = 2(\sin kx)(\cos \omega t)$

Now, we can compare with given equation.

23. 00000.16

Sol. $\ell = \ell_0 + \Delta\ell$, $r = r_0 + \Delta r$

By using formula for fundamental frequency.

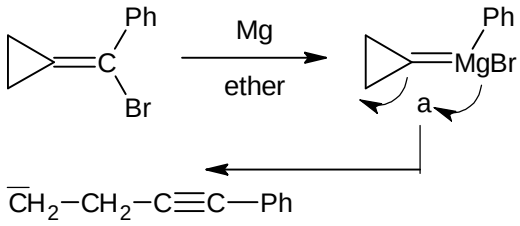
Chemistry

PART – II

SECTION – A

24. B

Sol.



$\overline{C}H_2-CH_2-C\equiv C-Ph$

i) H—CHO
ii) H^+

$Ph-C\equiv C-CH_2-CH_2CH_2-OH$ (B)

25. A

Sol. Tetrad AOS is C_4 & diad is C_2

26. D

Sol. Yellow turbidity of Sulphur is obtained in all the cases.

27. D

Sol.

$$K_2SO_4.Al_2(SO_4)_3.24H_2O \longrightarrow 2K^+ + 2Al^{3+} + 4SO_4^{2-}$$

n factor = 8	1 mole	2 mol	2 mol	4 mole
	8 gm eq	2 gm eq	6 gm eq	8 gm eq
	1 gm eq	1 gm eq cations		1 gm eq

28. A

Sol.

$$E_4 - E_3 = -\frac{2.18\times10^{-18}\times9}{16} + \frac{2.18\times10^{-18}\times9}{9}$$

$$= 2.18\times10^{-18}\times\frac{7}{16} = 9.53\times10^{-19}$$

29. A, C, D

Sol. $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$

In (b) one carbon is extra.

30. A, C, D

Sol. O – nitrobenzoic acid is stronger than m, p nitro derivatives as well as benzoic acid o – nitrophenol is weaker acid than P – nitrophenol due to intramolecular H – bonding.

31. B

Sol. $\pi = iCRT$

$\pi = 3\times0.4\times0.0821\times298$

$= 29.358$ atm

As osmotic pressure is more than applied pressure so osmosis will occur.

32. B

Sol. $\underset{M}{Li} + N_2 \longrightarrow \underset{A}{Li_3N}$

$Li_3N + H_2O \longrightarrow \underset{B}{LiOH} + \underset{C}{NH_3} \uparrow$

$NH_3 + CaOCl_2 \longrightarrow \underset{(D)}{N_2} + \underset{E}{CaCl_2} + \underset{F}{H_2O}$

$\underset{D}{N_2} + CaC_2 \xrightarrow{1100^oC} \underset{G}{CaCN_2} + \underset{H}{C}$

$\underset{H}{C} + O_2 \longrightarrow CO_2$

$\underset{G}{CaCN_2} + H_2O \longrightarrow \underset{I}{\underset{C}{CaCO_3}} + \underset{J}{NH_4OH}$

33. C

Sol. $NH_4NO_3 \xrightarrow{\Delta} N_2 + O_2 + H_2O$

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

SECTION – C

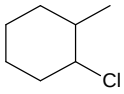
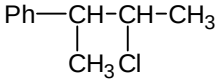
34. 3

Sol. $(A)\ is\ 2^0\ Alcohol$

$mol, wt\ of\ A = 60.$

$\therefore\ formula = C_3H_8o$

35. 2

Sol. In  &  rearrangement will give different products.

36. 5

Sol. $HI_n \rightleftharpoons H^+ + In$

$Color\ A \quad Colour\ B$

$k_{In} = \dfrac{\left[H^+\right]\left[In^-\right]}{\left[HI_n\right]}$

The midpoint of the colour change of an indicator HI_n is the point at which

$\left[I_n^-\right] = \left[HI_n\right]$

$\therefore\ ph = 5$

37. 6

Sol. $C_xH_y + O_2 \rightarrow Co_2 + H_2o$

$x = 100 \qquad = 300$

$x = 3$

38. 8

Sol. $T_c = 3.4.2k$

$= \frac{8a}{27Rb}$

$\frac{T_c}{P_c} = \frac{8b}{R}$

$b = 4N_A \times \frac{4}{3}\pi r^3$

$\therefore r = 1.62 \times 10^{-8}$ cm

$P_c = 72.9$ atm

$= \frac{a}{27b^2}$

39. 6

Sol. Let mol. wt of salt is M

(i) n-factor in acid – base titration = y

(ii) n – factor in redox titration = 2xz

Max of acid in 30 mL = Max of NaOH

$30 \times \frac{9.15}{M} \times y = 27 \times 0.12$ .....(i)

$30 \times \frac{9.15}{M} \times (27) = 36 \times 0.12$ .....(ii)

$\therefore \quad \frac{y}{z} = \frac{3}{2}$ .....(iii)

Total cationic charge = Total ammonic chare x + y = 2z

$\therefore$ x : y : z = 1 : 3 : 2

40. 1

Sol. $\frac{x}{x+16} + \frac{2x(2.18-x)}{2x+48} = 0.0187$

$x = 99 \simeq 100$

41. 2

Sol. Ma_3b_3 has two geometrical isomers i.e. fac & mer & both are optically inactive so x = 0

$M(AA)_3$ has one pair of enantiomers

$y = 1$

& Z = 1

so x + y + z = 2

42. 5

Sol. $Ag + O_2 + H_2S \longrightarrow Ag_2S + H_2O$

$Ag + H_2S \longrightarrow Ag_2S + H_2$

HgS,PbS & CuS are black

cds is yellow.

43. 8

Sol. $HCl + O_3 \longrightarrow Cl_2 + H_2O + O_2$

$PbS + O_3 \longrightarrow PbSO_4 + O_2$

$As + O_3 + H_2O \longrightarrow H_3AsO_4 + O_2$

$I_2 + O_3 \longrightarrow I_4O_9 + O_2$

$O_3 + K_2MnO_4 + H_2O \longrightarrow KMnO_4 + O_2 + KOH$

$Hg + O_3 \longrightarrow Hg_2O + O_2$

$O_3 + SO_2 \longrightarrow SO_3$

$O_3 + KI \longrightarrow I_2 + O_2 + KOH$

SECTION – D

44. 00087.50

Sol. % Enantomomeric excess = $\frac{30}{40} \times 100 = 75$

So % of d = $75 + \frac{25}{2} = 87.5$

45. 00001.66

Sol. Nearest distance between two 'A' layers = $\sqrt{3}a$

$\sqrt{3}a = 8\sqrt{6} \Rightarrow a = 8\sqrt{2}$ nm

$a = 8\sqrt{2}\ r \Rightarrow r = \frac{8\sqrt{2}}{2\sqrt{2}} = 4$

$\frac{r_{v.}}{r} = 0.414$

$r_{o.v} = 0.414 \times 4$

$= 1.656$ nm

46. 00007.00

Mathematics

PART – III

SECTION – A

47. B

Sol. $P_n = \frac{1+a_1}{a_1} \times \frac{1+a_2}{a_2} \times \ldots\ldots\ldots \frac{1+a_n}{a_n}$

$= \frac{a_2}{2 \times a_1} \times \frac{a_3}{3 \times a_2} \times \ldots\ldots\ldots \times \frac{a_n}{n \times a_{n-1}} \times \frac{a_{n+1}}{(n+1)a_n}$

$P_n = \frac{a_{n+1}}{\lfloor n+1} = \frac{(n+1)(a_n+1)}{(n+1)\lfloor n}$

$= \frac{1}{\lfloor n} + \frac{a_n}{\lfloor n} = \frac{1}{\lfloor n} + \frac{n(1+a_{n-1})}{n\lfloor n-1}$

$= \frac{1}{\lfloor n} + \frac{1}{\lfloor n-1} + \ldots\ldots\ldots\ldots\ldots + \frac{(1+a_1)}{\lfloor 1}$

$= 1 + \frac{1}{\lfloor 1} + \frac{1}{\lfloor 2} + \ldots\ldots\ldots + \frac{1}{\lfloor n}$

$\lim_{n \to \infty} P_n = e$

48. D

Sol. $\frac{\alpha}{r^3}, \frac{\alpha}{r}, \alpha r, \alpha r^3$ are terms

Equation – 1 $\frac{\alpha}{r^3} + \frac{\alpha}{r} + \alpha r + \alpha r^3 = -a$

$\alpha\left(r + \frac{1}{r} + r^3 + \frac{1}{r^3}\right) = a \qquad \ldots\ldots\ldots\ldots (A)$

Equation – 2 $\alpha^4 = d \qquad \ldots\ldots\ldots\ldots (B)$

Equation – 3 $\frac{\alpha^2}{r^4} + \frac{\alpha^2}{r^2} + \alpha^2 + \alpha^2 + \alpha^2 r^2 + \alpha^2 r^4 = b$

Equation – 4 $\frac{\alpha^3}{r^2} + \alpha^3 r + \alpha^3 r^3 + \frac{\alpha^3}{r} = -C \qquad \ldots\ldots\ldots\ldots (C)$

$(A)(C)$

$\frac{1}{\alpha^2} = \frac{-a}{c} \Rightarrow = \frac{c^2}{a^2}$

$d = c^2 / a^2$

$c^2 = da^c$

49. C

Sol. $\alpha = f(x)\big|_{-1}^{0} + f(x)\big|_{0}^{1}$

$= 0 - \frac{1/3}{1+\frac{1}{3}} + \frac{3}{1+3} - \frac{1}{1+0}$

$= -\frac{1}{4} + \frac{3}{4} - 1 = \frac{2}{4} - 1 = \frac{1}{2} - 1 = -1/2$

$\beta = \int_{-1}^{1} x - f(x).dx = xf'(x)\int_{-1}^{1} - \int_{-1}^{1} f'(x)$

$= 1 \times f'(1) - 0 + 0 - \frac{1}{4} + 1/2$

$= \frac{1}{2} + 1/2 = 1$

50. A, B, D

Sol. $\because \Delta SAB$ is right angle at S

Centre C is $\left(\frac{1+7}{2}, \frac{1-6}{2}\right) = (4, -3)$

$\therefore \alpha = 4$ and $\beta = -3$

$\therefore \alpha + \beta = 1$ and $\alpha - \beta = 7$

SC is radius and AB is diameter

$\therefore SC = \frac{1}{2}\sqrt{36+49} = \frac{1}{2}\sqrt{85} = \sqrt{\frac{85}{4}} = \sqrt{21.25}$

51. A, B, D

Sol. Confocalconics always intersect orthogonally & circle touches at vertex of ellipse.

$\therefore k = 1$

52. A, D

Sol. $y^3 - 3x^2y + x^3 = 0$

$\Rightarrow \frac{dy}{dx} = \frac{-\left(-6xy + 3x^2\right)}{\left(3y^2 - 3x^2\right)} = \frac{-3\left(-2xy + x^2\right)}{3\left(y^2 - x^2\right)} = \frac{y}{x}$

$\Rightarrow \int\left(\frac{dy}{dx}\right)dx$ i.e., $\int \frac{y}{x}dx = y + c$

$= f(x) + c$ or $xf'(x) + c$

53. A, B, C, D

Sol. $\left|2A^2 - 2A\right| = 144$

$2^2|A||A - I| = 144$

$\Rightarrow \alpha^2(\alpha - 1)^2 = 36$

$\alpha(\alpha - 1) = 6$ or -6

$\alpha^2 - \alpha - 6 = 0$ or $\alpha^2 - \alpha + 6 = 0$

$(\alpha - 3)(\alpha + 2) = 0$ or $\alpha = \frac{1 \pm \sqrt{1-24}}{2 \times 1}$

$\alpha = 3$ or -2 $\qquad = \frac{1}{2} \pm i\frac{\sqrt{23}}{2}$

$\therefore \text{tr}(A) = 2\alpha$

= 6 or –4 or 1 (Real part)

$|A| = \alpha^2 = 9$ or 4

54. A, B, D

Sol. $\because \begin{vmatrix} x & \sin x & 1+\sin x \\ 1 & \cos x & \cos x \\ 0 & \sin x & \sin x \end{vmatrix} = 0$

$\Rightarrow x = (2n+1)\frac{\pi}{2}$

55. C

Sol. $\because f(x) = \frac{4}{3}\sqrt{1-x^2} - \frac{\pi}{12}$

$\therefore$ range is $\left[-\frac{\pi}{12}, \frac{4}{3}, -\frac{\pi}{12}\right]$

56. A

SECTION – C

57. 4

Sol. $\sqrt{2(a^2+b^2)} = 4 \Rightarrow a^2+b^2 = 8$

$\therefore \frac{a^2+b^2}{2} \geq ab$

$\Rightarrow ab \leq 4$

58. 2

Sol. $\because f(x) = \sum_{i=1}^{\infty}\left[\frac{1}{(i-1)x+1} - \frac{1}{ix+1}\right]$

$\Rightarrow f(x) = \lim_{n\to\infty}\left[1 - \frac{1}{nx+1}\right]$

$\Rightarrow f(x) = 1$

$\therefore f\left(\frac{1}{x}\right) = 1$

$\Rightarrow f(x) + f\left(\frac{1}{x}\right) = 2$

59. 4

Sol. $\because \tan^{-1}x \geq \frac{2\pi}{3} - \frac{\pi}{2}$ for $-1 \leq 2-x^2 \leq 1$

$\Rightarrow \tan^{-1}x \geq \frac{\pi}{6}$ $\quad -3 \leq -x^2 \leq -1$

$x \geq \frac{1}{\sqrt{3}}$ $\quad 1 \leq |x| \leq \sqrt{3}$

$\therefore 1 \leq x \leq \sqrt{3}$

$\therefore$ have no solution for x < 1 or x > $\sqrt{3}$

So, $\alpha_{max} = 1$ and $\beta_{min} = \sqrt{3}$

60. 4

Sol. $I = \int_1^2 \frac{x^3 - 3x^2}{\ln(x-1)}$ and $J = \int_2^3 \frac{3x-5}{\ln(x-2)}$

$\Rightarrow J = \int_1^2 \frac{3x-2}{\ln(x-1)}$

$J + I = \int_1^2 \frac{x^3 - 3x^2 + 3x - 2}{\ln(x-1)} = \int_1^2 \frac{(x-1)^3 - 1}{\ln(x-1)} dx$

$= \int_1^2 \frac{(x-1)^3 - 1}{\ln(x-1)} dx = \int_0^1 \frac{x^3 - 1}{\ln x} dx = \ln 4$

$\therefore e^{\ln 4} = 4$

61. 2

Sol. $\overrightarrow{OA} = a\hat{i} + b\hat{j} + c\hat{k}$ and $a^2 + b^2 + c^2 = 4$

$\Rightarrow AB = \left(25 - 24\cos^2\frac{\theta}{2}\right)$

$\therefore |QD| = \frac{2}{5}\cos\frac{\theta}{2}$

$\Rightarrow 25|QD| = 2$

62. 4

Sol. $\frac{3x^2+1}{\sqrt{x^4+x^2}} = \frac{3x^2+1}{x\sqrt{x^2+1}} = \frac{x^2+1+2x^2}{x\sqrt{x^2+1}}$

$= \frac{\sqrt{x^2+1}}{x} + \frac{2x}{\sqrt{x^2+1}} \geq 2\sqrt{2}$

$f(x) = 2\sqrt{2}$

$\therefore x^2 + 1 = 2x^2$

$x^2 = 1$

$x = \pm 1 \qquad x = 1$

$\int_0^{\sqrt{2}} \frac{(3x^2+1)dx}{\sqrt{x^4+x^2}} \geq 2\sqrt{2} \times \sqrt{2} = 4$

63. 2

Sol. Given $a_{11} = b_{11}$, $a_{22} = b_{22}$, $a_{12} = -b_{12}$, $a_{21} = -b_{21}$

and $AB = BA = I$

$\Rightarrow A=\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{11} \end{pmatrix}$ and $|A|=1$

$\therefore a_{11}^2 - a_{12}a_{21} = 1$

$a_{12}a_{21} = a_{11}^2 - 1$

$\Rightarrow a_{12}$ or $a_{21} = (a_{11}-1)$ or $(a_{11}+1)$

$\Rightarrow |a_{12} - a_{21}| = 2$

64. 5

Sol. $f(x) = \dfrac{(\sqrt{5}-1)}{2x} \Rightarrow f(2) = \dfrac{\sqrt{5}-1}{4} = \sin\dfrac{\pi}{10}$

$\Rightarrow \sin^{-1}(f(2)) = \dfrac{\pi}{10}$

65. 5

Sol. $\because (r^2 - 4r + 2)S_r$

$\therefore \sum_{r=3}^{r=2011} \left|(r^2 - 4r + 2)S_r\right| == \dfrac{r-1}{\underline{|r-3}} - \dfrac{r}{\underline{|r-2}}$

66. 5

Sol. $\lim\limits_{x\to 0} \dfrac{a\tan x - b\sin x + cx^2 - x}{x^3} = \dfrac{3}{2}$

Then, $\dfrac{0}{0}$ form $\Rightarrow \lim\limits_{x\to 0} \dfrac{a\sec^2 x - b\cos x + 2cx - 1}{3x^2}$

Now, $\dfrac{0}{0}$ form means $a - b - 1 = 0$ …….. (1)

Again L'Hospital rule,

$\lim\limits_{x\to 0} \dfrac{2a\sec^2 x\tan x + b\sin x + 2c}{6x}$

$\dfrac{0}{0}$ form means c = 0

$\lim\limits_{x\to 0} \dfrac{2a\sec^2 x + 6a\tan^2 x\sec^2 x + b\cos x}{6}$

$\dfrac{0}{0}$ form means $\dfrac{2a+b}{6} = \dfrac{3}{2} \Rightarrow 2a + b = 9$ …….. (2)

$\therefore 3a = 10$ and $b = \dfrac{10}{3} - 1$

$a = \dfrac{10}{3}$ and $b = \dfrac{7}{3}$

$\therefore a + b + c = \dfrac{17}{3} = 5.66..... \Rightarrow [a + b + c] = 5$

SECTION – D

67. 00003.20

Sol. $\because 2\cos\theta_1 = \frac{8}{5} \Rightarrow \cos\theta_1 = \frac{4}{5} \Rightarrow \tan\frac{\theta_1}{2} = \sqrt{\frac{1-\frac{4}{5}}{1+\frac{4}{5}}} = \frac{1}{3}$ and $e = \frac{1}{2}$

$\therefore \tan\frac{\theta_2}{2} \times \frac{1}{3} = \frac{-\frac{1}{2}}{\frac{3}{2}}$

$\frac{\theta_2}{2} = \frac{3\pi}{4} \Rightarrow \theta_2 = \frac{3\pi}{2}$

$\therefore B = \left(0, -\sqrt{3}\right) \Rightarrow AB = 2 + 2 - \frac{1}{2} \times \frac{8}{5} = 4 - \frac{4}{5} = \frac{16}{5}$

68. 00540.00

Sol. $\because a_1^2 + b_1^2 + c_1^2 + a_2^2 + b_2^2 + c_2^2 + a_3^2 + b_3^2 + c_3^2 = 7$, where $\langle a_i, b_i, c_i \rangle$ are elements of ith row.

$\therefore$ Case I: All seven entries are 1 and rest of all zero

${}^9C_7 = {}^9C_2 = 36$

Case II: Only one entry is 2 & three entries are 1 and rest of all 0's.

${}^9C_4 \times {}^4C_1 = 504$

$\therefore$ Total = 540

69. 00000.75

Sol. Let $t = \log_{(3x+4)}(2x+1)$

$\Rightarrow \quad 2t + \frac{1}{t} = 3 \quad \Rightarrow \quad t = 1$ or $\frac{1}{2}$

$\log_{(3x+4)}(2x+1) = \frac{1}{2} \Rightarrow x = -1$ or $\frac{3}{4}$

$\therefore$ only solution is $x = \frac{3}{4}$